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Chemical Physics

Date: 1 July 2025

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Microscopic mechanisms of thermal transport at the SiO₂-water interface under the influence of wettability: A molecular dynamics study

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10.1016/j.chemphys.2025.112700 ↗

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Outline

[Highlights](#)[Abstract](#)[Graphical abstract](#)[Keywords](#)[1. Introduction](#)[2. Computational methodology](#)[3. Results and discussion](#)[4. Conclusion](#)

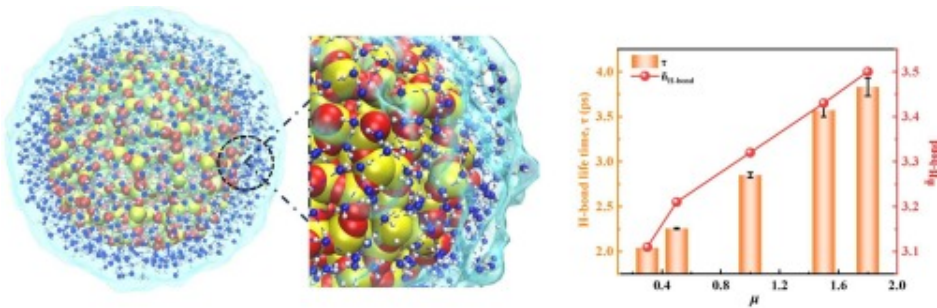
Highlights

- The TBC between the a-SiO₂ nanoparticles-water is 1.308 GW/m² K.
- The TBC increases with the enhancement of wettability.
- Solid-like a-SiO₂ nanoparticles interface behavior is key for TBC enhancement.
- Heat transfer mechanism is elucidated via H-bonds dynamics and VDOS analysis.

Abstract

Amorphous silica (a-SiO₂) nanoparticles play an important role in the fields of solar thermal utilization, nanofluids and thermomagnetic therapy. However, there is still a lack of understanding of the microscopic mechanism of heat transfer between nanoparticles and water, especially the wettability of nanoparticles. Here, we use molecular dynamics (MD) simulation to study the microscopic mechanism of heat transport between a-SiO₂ and water under the different wettability. The results show that the hydrogen bonds(H-bonds) heat conduction network is formed at the interface of the a-SiO₂ nanoparticles due to the H-bonds interaction between water molecules. In addition, with the increase of wettability, the thermal boundary conductance also increases. This is due to the formation of more H-bonds between water molecules at the interface of the a-SiO₂ nanoparticles, which slows down the exchange frequency of water molecules between the a-SiO₂ nanoparticles interface and the bulk water, increases the time scale of stay on the a-SiO₂ nanoparticles surface, and increases the probability of heat transfer between a-SiO₂ nanoparticles and water. The change of wettability leads to the change of water structure at the a-SiO₂ nanoparticles interface, forming a tetrahedral hydrogen bond structure similar to that of solid ice. The analysis of VDOS further confirms the key role of the vibration of the H-bonds network in the heat transport at the solid-liquid interface. Consequently, these insights provide a valuable theoretical framework for comprehending heat transport mechanisms at soft-hard interfaces.

Graphical abstract



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Funding sources

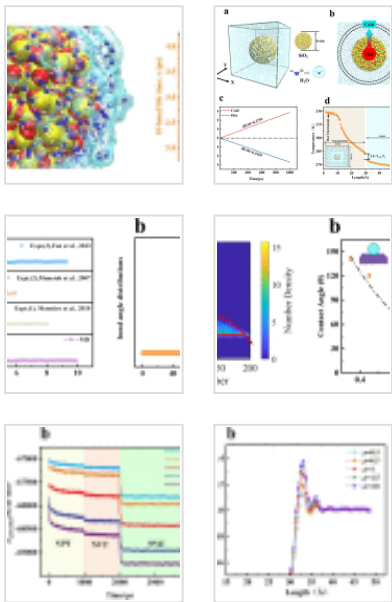
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References

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Figures (10)



[Show 4 more figures](#)

Tables (1)

Table 1. CWCA Parameters for Silica-Wa...

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Keywords

Nanoparticle-water interface; Hydrogen bonds; Wettability; Thermal boundary conductance; Vibrational density of states; Molecular dynamics

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1. Introduction

Heat transfer across solid-liquid interfaces is omnipresent in applications such as polymer composite materials, [1] nanofluids, [2] nanoparticle based cancer treatments, [3] batteries, [4] bioelectronic chips, [5,6] and micro/nano heat sinks. [7] When the characteristic size of the structure is close to the nanometer scale, the thermal resistance generated by the interface plays an important part in the whole heat transfer process, and even dominates. [8,9] Research related to heat transport across solid-liquid interfaces began with Kapitza's study of helium flow of ⁴He (Helium II) at low temperatures, where temperature discontinuities are observed as heat passes through the interface [8,10]. This mismatch gives rise to thermal boundary resistance (TBR) or Kapitza resistance and it can be defined as $q = \Delta T / R_k$, where q is the heat flux, ΔT is the temperature discontinuity at the interface, and R_k is the TBR. Alternatively, some researchers prefer to use of the thermal boundary resistance inverse, commonly known as the thermal boundary conductance (TBC) represented by G . [10,11]

At present, studies on TBC have found that not only the temperature discontinuity at the solid-liquid interface, but also the structure and dynamic characteristics of the liquid will change significantly. Liquid molecules are easily adsorbed on the solid surface to form an adsorbed water layer. This structure is vividly described as a bridge for interfacial heat

transport. [[12], [13], [14], [15]] At present, the research on the heat transport at the solid-liquid interface focuses on how to promote the liquid to form a stable adsorption layer on the solid surface. Researchers have induced a stable liquid adsorption layer at the solid interface by adjusting the interface temperature, [16,17] pressure, [18] roughness, [19,20] patterning, [21,22] and functionalization, [[23], [24], [25]] surface charge [26,27] and other factors. The thermal transport mechanisms at solid-liquid interfaces have been categorized into two types by Wei and colleagues [28]: interfacial bonding and vibrational matching, which are applicable for thermal management at such interfaces. For instance, functionalizing solid surfaces with self-assembled monolayers (SAMs) has revealed that interface heat transfer can be enhanced by bridging the vibrational modes of solids and liquids, promoting the formation of interfacial H-bonds, and increasing surface wettability. [[29], [30], [31]] Unquestionably, a deeper understanding of the thermal transport at solid-liquid interfaces will provide fresh insights into the further optimization and adjustment of solid surfaces. Furthermore, current research predominantly focuses on the vital role of the adsorbed water layer in the thermal transport at solid-liquid interfaces. However, as the characteristic scale diminishes to several nanometers, the thermal transport mechanism within the water film at nanoparticle interfaces may differ from the bulk properties. The H-bonds network of water molecules at nanoparticles interfaces is disrupted, the dynamics of water's rotational freedom slow down, and the dipoles of water molecules may alter compared to those in bulk water [[32], [33], [34]]. Therefore, the transformation of the water molecule structure within the adsorbed layer, the arrangement and orientation of water molecules, and the behavior of H-bonds dynamics will significantly impact the thermal transport at solid-liquid interfaces. The scarcity of prior research in these areas has spurred our investigation.

Although a plethora of experimental analyses have been conducted to investigate the thermal transport characteristics at the silica-water interface under various influencing factors, the underlying mechanisms remain elusive. For instance, Ali Alkurdi et al. [35] employed ultrafast laser pulses to compare the heat transfer at gold-water and silica-

water interfaces, with the experimental results indicating a higher interfacial heat transfer efficiency for the silica-water system, attributed to the enhanced electron-phonon coupling at their interface. Moreover, Hu et al. [36] conducted pump-probe experiments to compare the interfacial heat transfer in silica-coated gold nanoparticles in both water and ethanol solutions, revealing that the characteristic time constant for heat dissipation depends on the thickness of the silica shell and the solvent. These experimental findings suggest that the silica-water interface possesses unique advantages in terms of thermal transport performance. Nevertheless, at the nanoscale, experimental research encounters limitations in fully understanding the intrinsic mechanisms of interfacial heat transfer between water and silica, especially for silica surfaces with different wettability.

The MD simulations are a powerful computational tool for analyzing nanoscale phenomena at the molecular level, [[36], [37], [38], [39]] and they have been widely utilized to elucidate the interfacial heat transfer mechanisms in silica-water systems. Schoen et al. [40] employed MD simulations to investigate the heat transfer at hydrophilic silane-water interfaces and hydrophobic silane-water interfaces. Due to the role of silane-water H-bonds, the interfacial heat transfer efficiency was higher at the hydrophilic silane-water interface compared to the hydrophobic one. Additionally, they observed that the TBC at the hydrophilic silane-water interface decreased with increasing temperature, attributed to a reduction in the number of interfacial H-bonds. Hu et al. [41] studied the interfacial heat transfer between hydrophilic quartz and water, revealing that the overall TBC increased significantly with the thickness of the water layer, and this increase exhibited a non-intuitive, non-monotonic dependence on the water layer thickness. Sun et al. [42] explored the influence of surface ionization on the interfacial heat transfer between silica and water using MD simulations. Their results indicated that increasing the surface ionization significantly improved TBC, as it led to the formation of more H-bonds between ionized silica and water, thereby creating more effective pathways for interfacial heat transfer. Hu et al. [42] utilized non-equilibrium molecular dynamics (NEMD) to modify silica surfaces with SAMs, revealing the thermal rectification effect at

the silica-SAMs-water interface. Sun et al. [43] studied the interfacial heat transfer between five different silica surfaces and two organic liquids using MD simulations, demonstrating that an increase in the density of silane hydroxyl groups enhanced the interfacial heat transfer in the silica–triacontanol system, albeit with minimal effect on the silica–triacontanol system. Xu et al. [44] employed MD simulations to investigate the interfacial heat transfer between water and silica with varying surface densities of silane hydroxyl groups, emphasizing the competitive nature of water-silane and silane-silane H-bonds in influencing interfacial heat transfer and proposing the use of an effective hydroxyl density to characterize the interfacial heat transfer capability of silica-water interfaces. However, to the best of our knowledge, the influence of different wettability surfaces on interfacial heat transfer between silica and water has been rarely considered. In fact, silica surface wettability is unavoidable in practical applications and can be modulated by grafting functional groups, surface modification, and adjusting surface ionization. Therefore, the adjustable silica surface wettability brings about complex and variable surface properties, significantly impacting interfacial heat transfer. Consequently, exploring the impact of different wettability silica and water interfacial heat transfer has practical significance, and the scarcity of previous research has prompted our investigation.

This study employs MD simulation techniques to investigate the microscopic mechanisms of heat transport between the a-SiO₂ nanoparticles and water. It focuses on elucidating the influence of wettability on the thermal transport across a-SiO₂ nanoparticles and water interfaces, delving into a detailed examination of the H-bonds lifetime, average number of the H-bonds, and the H-bonds autocorrelation functions within the a-SiO₂ nanoparticle's adsorbed layer to reveal the dynamic behavior of water molecules at the a-SiO₂ nanoparticles interface. Additionally, by calculating the vibrational density of states, we gain insights into the microscopic vibrational information, further corroborating the pivotal role of H-bonds stretching vibrations in interfacial heat transport. The relationship between the hydrogen bonds dynamics of the adsorbed water layer at the a-SiO₂ nanoparticles interface and solid-liquid interfacial transport is the primary objective of

this research. The article has the following structure. Section 2 provides a detailed account of the computational aspects of the simulations, Section 3 discusses the hydrogen bonds dynamics of water molecules at the particle interface, and Section 4 concludes with a summary of the findings and final remarks.

2. Computational methodology

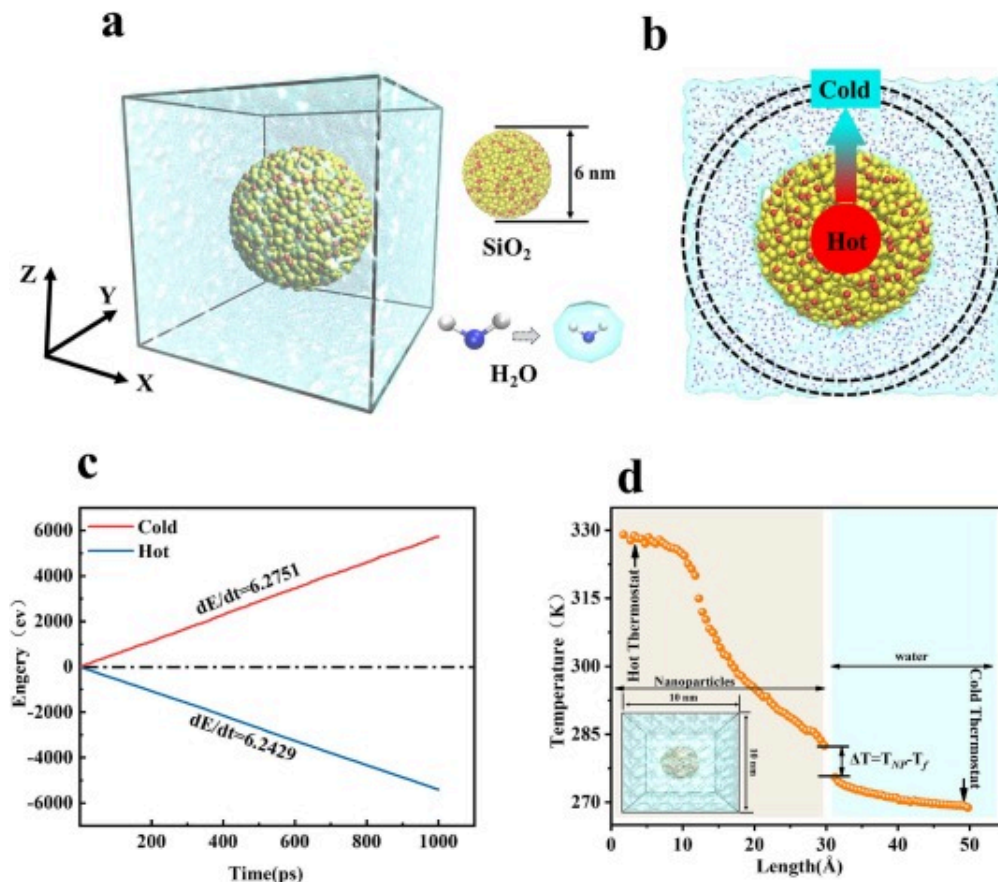
2.1. Preparation of amorphous silica nanoparticles

Amorphous silica nanoparticles (a-SiO₂) were fabricated through the molten-quench method [45,46] from full-density crystalline silica, [46] a process that includes initialization, heating to an ultra-high temperature, followed by quenching to ensure the desired amorphous state. Thereafter, spherical a-SiO₂ nanoparticles with a diameter of 60Å were meticulously excised from the bulk amorphous silica. [47] The radial distribution function (RDF) and angular distribution function (ADF) of the resultant a-SiO₂ cubes were compared with reference data to ascertain the precise formation of a-SiO₂.

2.2. Interfacial heat transfer modeling

2.2.1. Modeling and simulation details

The model constructed for this study, as depicted in [Fig. 1](#), features a 60Å diameter a-SiO₂ nanoparticles placed within a simulation box (100Å×100Å×100Å). Water molecules were modeled at a density of 998kg/m³, corresponding to 293K, and were randomly distributed around the a-SiO₂ nanoparticles using the Packmol software, with a total of 83,037 water molecules. [48] Throughout the simulation, the density of water and the position of its center of mass were maintained constant.



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Fig. 1. (a) Schematic diagram of the a-SiO₂ nanoparticles-water system; (b) schematic diagram of the a-SiO₂ nanoparticles-water interface heat transfer model. For a-SiO₂ nanoparticles-water interface heat transfer typical (c) heat flux and (d) temperature gradient

The wettability of SiO₂ is determined by the density of silanol groups on its surface, a question that has been the subject of in-depth investigation. [45,49] For instance, Cruz-Chu and colleagues [49] have explored the influence of silanol groups on wetting behavior. Xu and colleagues [50] have examined the interfacial heat transport issues

between hydroxylated silica surfaces and water. Federico and others [51] have investigated whether water adsorption disrupts the hydrogen bonds pattern of the pristine free silica silanol surface and the extent to which the disturbances caused by surface contact affect the multilayer structure of water. In this study, the molecular modeling of the α -SiO₂ employs complete dehydroxylation; however, this does not imply that the role of silanol groups is overlooked. Instead, their influence is accounted for through the introduction of a fluid-solid interaction strength parameter. To adjust surface wettability, different solid-fluid interactions are established by tuning the values of the energy parameters: $\epsilon^* = \mu \cdot \epsilon$, where μ is a coefficient ranging from 0.3 to 1.8. This approach serves to reduce the complexity of molecular modeling and has been widely utilized in the study of thermal fluid engineering. [52]

The water molecules were modeled using a rigid TIP3P model, which incorporates a short-range Lennard-Jones potential and a long-range Coulombic term. [53] The SHAKE algorithm was employed to maintain the rigidity of the bonds (O–H) and angles (H–O–H), treating them as harmonic springs with coefficients of 19.51 eVÅ⁻² and 2.4 eVrad⁻², respectively. The PPPM algorithm was utilized to handle the long-range Coulombic terms with a precision of 1×10^{-6} , and the cutoff distance for Coulombic interactions and Lennard-Jones potential was set at 10 Å. [47]

The fluid–solid interactions between water and silica are described by the CHARMM water contact angle (CWCA) force field, proposed by Cruz-Chu et al., [49] which is grounded in short-range Lennard-Jones potential (Eq. 1) and the long-range coulombic terms (Eq. 2). The U_{vdW} denotes the van der Waals (VdW) term; U_{Coul} signifies the electrostatic coulombic term; r_{ij} represents the distance between atoms i and j ; and ϵ_0 is the vacuum permittivity with 0.15 kcal/mol. In silica, the charge of Si atom is 0.900 e and that of O atom is -0.450 e. [54] In the TIP3P water model, the charge of the O atom is -0.834 e, and that of the H atom is 0.417 e. [53] Interactions between different atoms are determined using the Lorentz-Berthelot combination rules.

$$U_{\text{vdW}} = \sum_i \sum_{j>1} \varepsilon_{ij} \frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^6}{r_{ij}^6} \quad (1)$$

$$U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \quad (2)$$

The force field has been extensively utilized in simulating water-silica interactions, [52] with its meticulous parameters detailed in Table 1. The CHARMM force field is grounded in the rigid TIP3P water model and amorphous silica, where the parameters within the force field have been optimized, including the explicit charges of silica and its interaction with the charges in water, to accommodate the experimental measurements of the water-silica contact angle. Consequently, this force field can be employed to measure the water contact angle on the surface of an amorphous silica plates, thereby ascertaining the wettability of the amorphous silica nanoparticles.

Table 1. CWCA Parameters for Silica–Water Interactions.

Pair	ε (meV)	σ (Å)	q (e)
H in water	1.995	0.4	0.417
O in water	6.596	3.1506	−0.834
Si in silica	13.01	3.8264	0.9
O in silica	6.596	3.118	−0.45
Angle	ε (eVrad ^{−2})	σ (Å)	
H-O-H	2.40	104.52	
Bond	ε (eVÅ ^{−2})	σ (Å)	
O-H	19.51	0.9572	

2.2.2. Calculation of thermal boundary conductance

Utilizing the NEMD approach to compute the TBC, the computational formula is illustrated as Eq. (3) [14].

$$\text{TBC} = \frac{J}{A\Delta T} \quad (3)$$

where ΔT represents the temperature difference at the interface between the a-SiO₂ nanoparticles and water, A is the surface area of the a-SiO₂ nanoparticles, given by $A = 4\pi R^2$, and J is the heat flux across the a-SiO₂ nanoparticles interface, the calculation of which is presented in Eq. (4).

$$J = \frac{1}{2} \frac{Q_h + Q_c}{\Delta t} \quad (4)$$

where Q_h and Q_c denote the cumulative energy within the cold and hot regions, respectively, over the operational duration Δt . The heat flux, J , is determined through the virial stress tensor formulation, and within the LAMMPS program, this method is restricted to pairwise potentials. However, since the potential function for SiO₂ employs the Tersoff many-body potential, the present computation has been modified in accordance with the method proposed by Matsubara and colleagues. [55] The nanoparticles is divided into two segments, core and shell, based on atomic positional coordinates: $0 < r < 1.2 \text{ nm}$ and $1.2 \text{ nm} < r < R$. The atoms in the core region are the heat source that produces a steady stream of heat, while the water molecules in the outermost region are the source of cold. This division is necessitated because employing the Langevin thermostat to control temperature results in the emergence of a “spurious interfacial temperature jump”. [[56], [57], [58]] The schematic of the computational model is depicted in Fig. 1(b).

2.2.3. Calculation of vibrational dynamic density

In order to reveal the potential mechanism of thermal boundary conductance enhancement at the silica-water interface, the corresponding phonon information of atoms in the system is studied. The vibrational density of states (VDOS) is calculated here. The VDOS is used to describe the intensity of phonon vibration modes at different

frequencies and is often used to analyze the causes of thermal transport changes. The VDOS can be calculated by the velocity autocorrelation function (VACF) fast Fourier transform: [59].

$$VDSv = \int \gamma t \exp -2\pi i v t dt \quad (5)$$

Where v denotes frequency, i represents the imaginary unit, and γ signifies the velocity autocorrelation function (VACF), which can be derived from the equation

$\gamma t = \sum_i v_{i0} \cdot v_{it} / \sum_i v_{i0} \cdot v_{i0}$, where $v_i(0)$ and $v_i(t)$ are the initial velocity of an atom and its velocity at time t , respectively. During computation, the VACF is calculated after the simulation has run for over 10ps, at which point the structure has reached an equilibrium state.

2.2.4. Interface liquid structure

In addition, the local RDF at the a-SiO₂ nanoparticles interface was determined to delineate the structural changes of water molecules at this juncture. The domain for this determination spans from the nanoparticles to the nadir of the first density valley, as illustrated by the RDF calculation Eq. (6):

$$gr = nr \frac{nr}{4\pi r^2 \rho \Delta r} \quad (6)$$

Where $n(r)$ represents the number of target atoms within the shell between r and $r+\Delta r$, distant from the central atom, ρ denotes the average density of the target atoms, and r is the inner radius of the shell.

The survival probability of water molecules within the interfacial layer was computed to ascertain the temporal scale of their presence at the particle interface; the calculation is expressed in Eq.7: [60].

$$St = \frac{Nt + \tau}{Nt} \quad (7)$$

Where N denotes the number of water molecules, t represents the time step, and the angular brackets signify the average over all initial times.

2.2.5. Simulation details

All simulations were conducted using the open-source molecular dynamics package LAMMPS, [61] with visualization of the results facilitated by the VMD software. [62] The computational procedure was executed as follows: (1) Firstly, velocities were scaled to a temperature of 5K for initial velocity initialization; energy minimization was performed to eliminate artificially induced strains; (2) Secondly, all simulation systems were thermalized in the NPT ensemble at the pressure of 1 atm and the temperature of 300K, allowing water molecules to achieve the correct density over a duration of 1 ns; (3) Thirdly, the temperature was kept at 300K using the Nosé-Hoover thermostat (NVT ensemble) over a period of 1 ns, which enabled the system to relax appropriately at the target temperature. (5) Finally, thermal transport simulations were conducted in the NVE ensemble for data collection. To facilitate NEMD simulations, regions designated as heat source and cold source were demarcated, with the temperatures in the cold and hot regions were set as 270K and 330K, respectively. The entire simulation box was divided into 100 concentric rings along the radial direction of the nanoparticles to record the average temperature of each ring, thereby generating a temperature gradient. The heat transfer simulation spanned 2 ns, with the first 1 ns dedicated to achieving stable temperature and heat flow distributions, while the last 1 ns of MD simulation was averaged to obtain the temperature gradient and heat flux, mitigating the impact of system randomness on the accuracy of the simulation data. Fig. 1(c) illustrates a typical temperature distribution across the α -SiO₂ nanoparticles-water interface. The corresponding heat flux J is recorded by the average of the input heat flux in the cold region and the output heat flux in the hot region by Eq. (4). Fig. 1(d) depicts a typical heat flux density curve, with the slope indicating structural stability, allowing for the calculation of the TBC via Eq. 3. Periodic boundary conditions were applied in all three dimensions to negate boundary effects, and a time step of $\Delta t = 1$ fs was selected. The

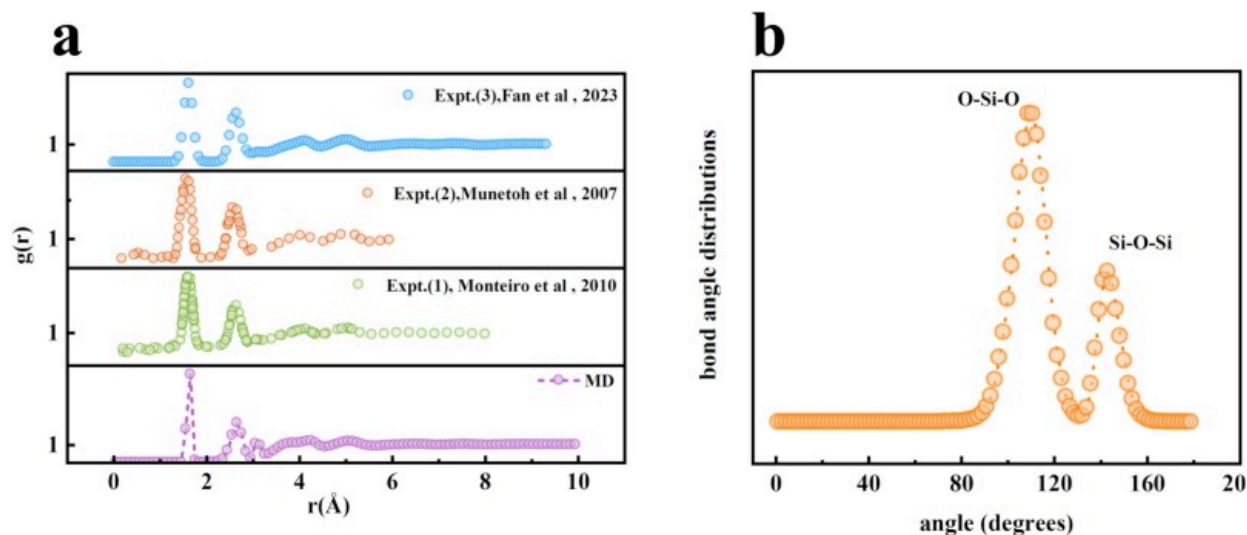
positions and velocities of each atom were updated using the standard Velocity-Verlet algorithm, [63] with simulation data output every 1000 time steps.

3. Results and discussion

3.1. Validation of model

3.1.1. Structure of a-SiO₂ nanoparticles

The a-SiO₂ nanoparticles model employed an enhanced Tersoff potential, which accurately captures the thermodynamic properties of the a-SiO₂; the specific functional form and additional details are referenced in this literature. [64] To characterize the short-range order of the a-SiO₂ structure, the RDF was calculated at a temperature of 300K. As depicted in Fig. 2(a), the MD simulation results exhibit remarkable consistency with both ab initio calculations and experimental measurements in terms of peak positions and heights. [[64], [65], [66]] Through the computation of the ADF for a-SiO₂, it was determined that the Si–O–Si bond angle distribution's prominent peak is located at 110.7°, and the highest peak in the Si–O–Si bond angle distribution is situated at 143.10°, in close agreement with ab initio data. [64,67]



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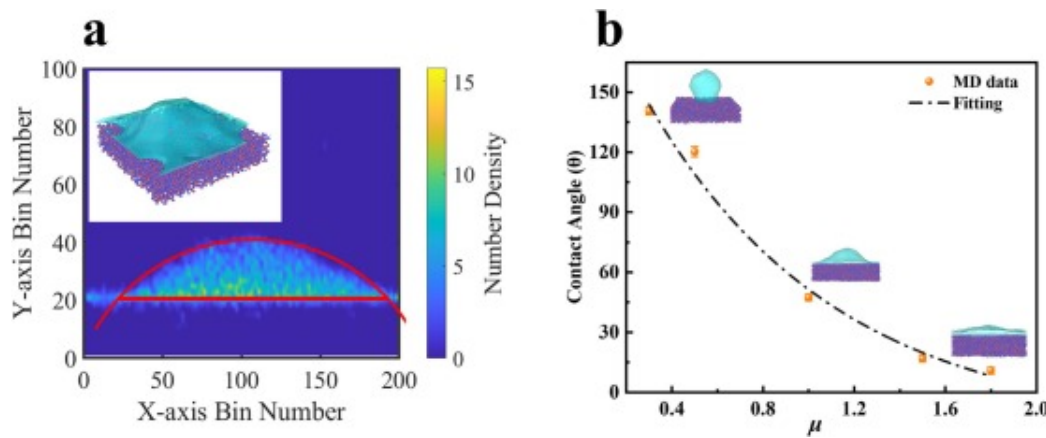
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Fig. 2. (a) The RDF of a-SiO₂ nanoparticles calculated by MD is compared with the reference; (b) the ADF of a-SiO₂ nanoparticles calculated by MD.

3.1.2. Verification of interface heat transfer model

By modulating the strength of the interactions between water molecules and the a-SiO₂ nanoparticles, the tunability of the a-SiO₂ nanoparticles surface wettability can be achieved. Within MD simulations, the interaction between silica and water is adjusted by setting the interaction parameter μ , thereby generating varying wettability outcomes. However, μ is not a parameter adjustable in experimental settings, hence the use of contact angles to characterize wettability and elucidate the significance of μ . The precise simulation process for contact angles during MD is as follows: the silica plate is fixed in the system, and water molecules undergo relaxation at 300K in the NVT ensemble, with the simulation running for 2ns and a time step of 0.5fs. The first 1ns is used to achieve equilibrium between the droplet and the substrate, and the final 1ns is used for data

collection. With the density map of the droplet on the surface and the method for calculating the contact angle illustrated in Fig. 3(a).



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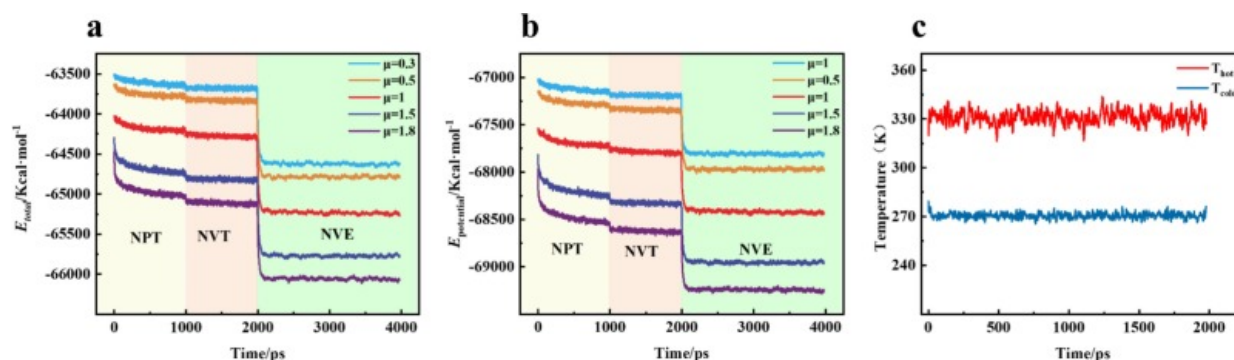
Fig. 3. (a) Calculation method of contact angle of droplets on a-SiO₂ substrate and droplet profile; (b) correlation between contact angle and μ .

The dependence of the contact angle on μ is depicted in Fig. 3(b). It is evident that the CWCA force field employed in this study can accurately describe and predict the contact angle of water on a silica surface. For instance, for a fully dehydroxylated silica surface ($\mu=1$), this work yields a contact angle of 47°, which is in close agreement with experimental values (42°) and predicted values (48°) reported in the literature. [52,68] The results of the MD simulations are mathematically fitted as follows:

$$\theta = 232.85443 \exp -\mu / 0.88859 - 27.52941 \quad (8)$$

To ensure the system's stability prior to the calculation of the TBC, this study concurrently monitored the variations in the system's total energy, potential energy, and the temperatures of the heat and cold regions during the temperature control phase, as illustrated in Fig. 4(a-c). The results revealed that under all conditions, neither energy nor

temperature exhibited significant fluctuations during the temperature control phase, signifying that the system had achieved a stable state. Moreover, to enhance data accuracy, only the data from the last 1 ns of the temperature control phase was utilized for the calculation of the TBC, thereby reducing the errors introduced by the randomness of the temperature control algorithm. Multiple calculations were performed, and their average value was taken to further ensure the reliability and accuracy of the data.



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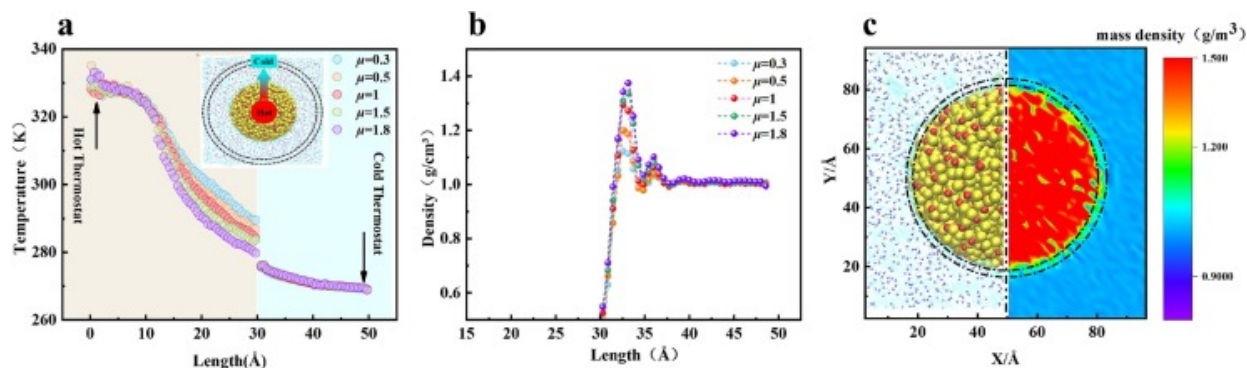
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Fig. 4. System (a) total energy and (b) potential energy with (c) cold source and heat source temperature changes.

3.2. The relationship between TBC and wettability

The temperature distribution between the a-SiO₂ nanoparticles and water is illustrated in Fig. 5(a), revealing several salient features. Firstly, at the interface of the a-SiO₂ nanoparticles, a conspicuous temperature drop is observed at the juncture between the region influenced by the thermostat and the region unaffected by it. The temperature discontinuity at this interface is attributed to the phonon mismatch resulting from the thermostat's dynamic rescaling. This artifactual effect of the thermostat has been previously investigated in the literature, [[56], [57], [58]] and the temperature gradient produced within the thermostat's domain differs from that in the thermostat-free region, contingent upon the system's temperature and heat flux, as meticulously studied. [58]

The a-SiO₂ particle region exhibits a temperature profile shaped by the interplay of silica phonons. Temperature within the a-SiO₂ nanoparticle region exhibits a declining trend as it moves away from the temperature-controlled area. However, this decline slows down in proximity to the water-contacting region, a phenomenon attributed to phonon scattering at the interface.



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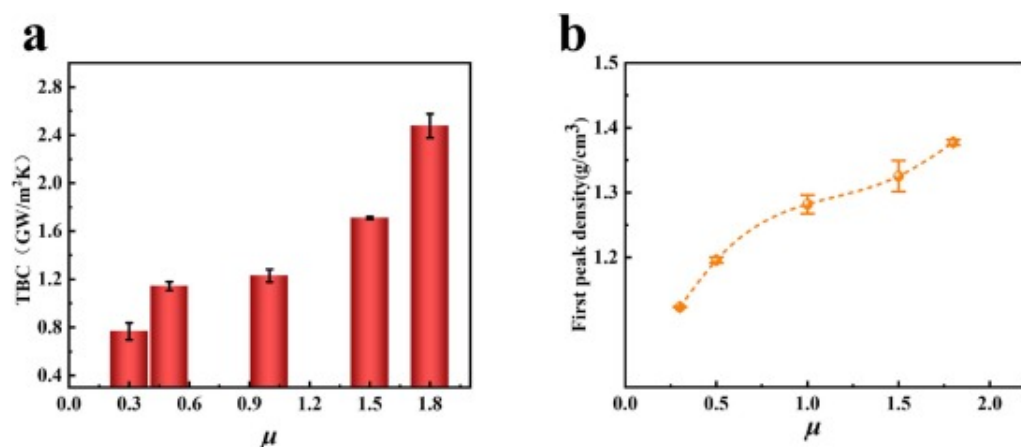
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Fig. 5. The interface of a-SiO₂ nanoparticles under different wettability (a) temperature jump and (b) one-dimensional density line diagram of water molecules; (c) two-dimensional density cloud of water molecules at the interface of a-SiO₂ nanoparticles.

Moreover, it is distinctly apparent at the nanoparticles interface that the liquid temperature adjacent to the solid material experiences greater fluctuations than that in the main body of the liquid, a temperature variation that diminishes with increasing wettability. Oscar employs a generalized acoustic model to elucidate this phenomenon, where the discordance of atomic vibrations between the two media precipitates a temperature jump at the interface. [69]

The TBC between a-SiO₂ nanoparticles and water is calculated using Eq. (3), with the interface temperature difference and heat flux, as shown in Fig. 6(a). The figure illustrates

that as wettability increases, the temperature jump at the interface diminishes, and correspondingly, the numerical value of the TBC increases.



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Fig. 6. Interface characteristics of a-SiO₂ nanoparticles under varying wettability conditions: (a) the TBC; (b) trend of the first density peak variation in water.

In addition to the obvious temperature jump at the nanoparticles interface, it is also found that the density of water at the nanoparticles interface fluctuates significantly, and there is a peak at a certain position from the nanoparticles interface. Relevant studies have shown that the interface thermal conductance is closely related to the density of the fluid at the nanoparticles interface. To further elucidate the relationship between wettability and interface temperature variations, the trends in water density near the nanoparticles interface under different wettability were calculated, as illustrated in Fig. 5(b). The water density distribution curve at the nanoparticles interface exhibits an oscillatory trend, featuring two density peaks, with the first peak being larger than the second and increasing in size as wettability increases. Away from the nanoparticles, the water density distribution reverts to a state resembling “bulk” water. The interaction between water molecules and the nanoparticles results in the formation of an adsorption

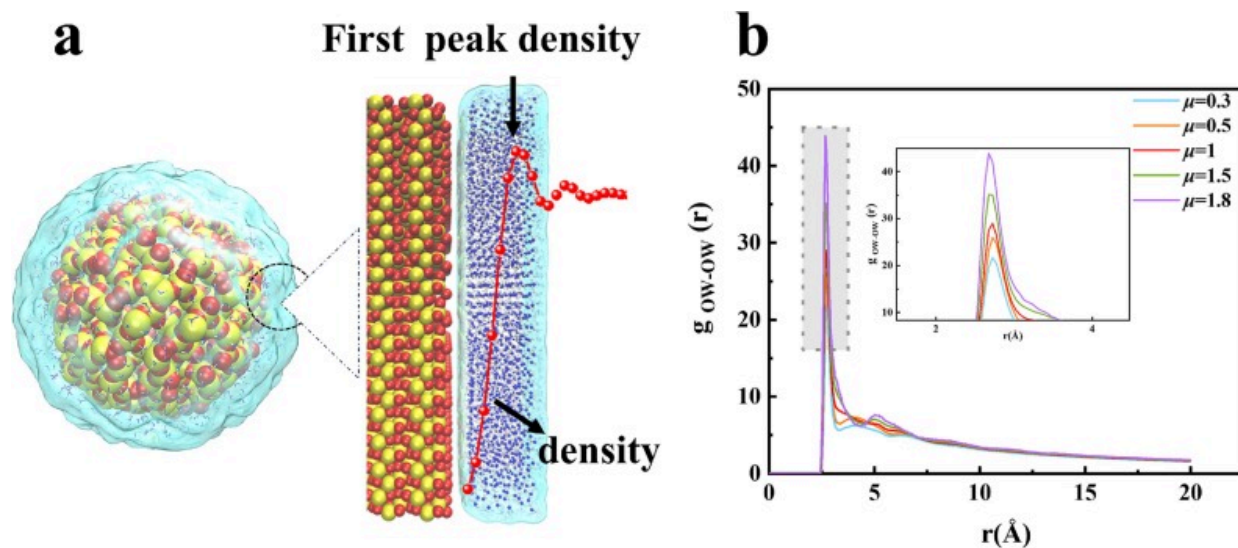
layer near the nanoparticles. Additionally, significant differences in water density distribution were found between “strong wettability ($\mu=1.8$)” and “weak wettability ($\mu=0.3$)”. Under “strong wettability” the water molecules at the nanoparticles interface exhibit a higher density peak, indicating a greater number of water molecules are adsorbed at the nanoparticles interface, forming a dense structure around the nanoparticles. This layering effect, induced by the nanoparticles, extends into the “bulk” water, as confirmed by the two-dimensional density cloud map in Fig. 5(c). At the nanoparticles interface, the surface forces and liquid-liquid interactions at the nanoscale result in the solid-state freezing of atoms near the solid surface, forming a “solid-like liquid layer” as termed by Xue et al. [12] The water molecules in these layers exhibit more order than those in the majority of the liquid, presenting features reminiscent of a crystalline structure. The presence of this unique double-density adsorption peak, or the “solid-like liquid layer” directly contributes to the enhanced thermal transport at the solid-liquid interface. This, in turn, explains the gradual increase in interface thermal conductance as wettability increases.

Furthermore, as wettability increases, the first density peak also grows in magnitude, and the position of the first density peak is typically employed to delineate the effective thickness of the “solid-like liquid layer”. To investigate the correlation between TBC and the first density peak, Fig. 6(b) depicts the water density at the first density peak of nanoparticles under different wettability in a stable heat flux condition. The results indicate that the numerical value of the first density peak increases with increasing wettability. A higher density peak signifies a greater number of water molecules adsorbed at the nanoparticles interface; however, we observe from Fig. 5(b) that the position of the first density peak does not undergo a significant shift, indicating that the thickness of the adsorbed layer does not become thicker with increasing wettability.

3.3. H-bond dynamics analysis of water molecules at solid-liquid interface

By investigating the structural changes inside the “solid-like liquid layer”, the promoting effect of enhancing the interface thermal conductance is further revealed. [23,36] The

RDF of the “solid-like liquid layer” is calculated to delineate the structural changes of water molecules at the nanoparticles interface. Within MD simulations, the SHAKE algorithm is utilized to constrain the bonds and angles of water molecules, treating them as rigid bodies; hence, the position of the oxygen atom can be considered representative of the water molecule's position. In the RDF calculation, oxygen atoms are employed to represent water molecules, and it is reasonable to characterize the distribution of water molecules using the distribution of oxygen atoms. Moreover, the water molecules from the first density trough to the nanoparticles interface are selected as the “solid-like liquid layer” to calculate the RDF, with the schematic of the calculation region depicted in Fig. 7(a). The RDF calculation results are presented in Fig. 7(b), with the increment of r , the value of the $g(r)$ function gradually diminishes and converges to 1.0, indicating that water molecules exhibit a microstructural characteristic of short-range order and long-range disorder, in accordance with the arrangement rules of liquid molecules. Moreover, the $g(r)$ function clearly exhibits two peaks, which align with experimental results. [32] The primary hydration peak resides at the same position of 0.28 nm, primarily due to H-bonds formed between water molecules within the first hydration shell. The higher the first peak of $g(r)$, the stronger the H-bonds interaction and the stronger the binding with surrounding water molecules. As wettability increases, the hydration peak at 0.28 nm intensifies, indicating stronger solid-liquid interactions, a higher probability of water molecules adsorbing to the α -SiO₂ nanoparticles interface, and the formation of more H-bonds among water molecules, resulting in a more stable and dense “solid-like liquid layer” that readily establishes heat transfer channels at the interface, thereby enhancing the probability of heat transfer and leading to a greater interface thermal conductance. Concurrently, the appearance of a second absorption peak at 0.45 nm and its increase with wettability suggest that the long-range order of water molecules is gradually increasing, indicating a transition in the water structure to a state intermediate between liquid and solid.



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Fig. 7. a-SiO₂ nanoparticles interface “solid-like liquid layer” (a) schematic diagram (b) RDF of water molecules.

Concurrently, the presence of higher peaks signifies that water molecules are more interconnected via H-bonds with adjacent molecules. By analyzing the kinetic behavior of H-bonds at the nanoparticles interface, the microscopic mechanisms underlying the dynamic structural changes of the “solid-like liquid layer” and interfacial thermal transport are further elucidated. H-bonds are essentially covalent bonds between H atoms and more electronegative atoms or groups (donors, D) that attract an additional lone pair electron (acceptors, A) electrostatically. [70]

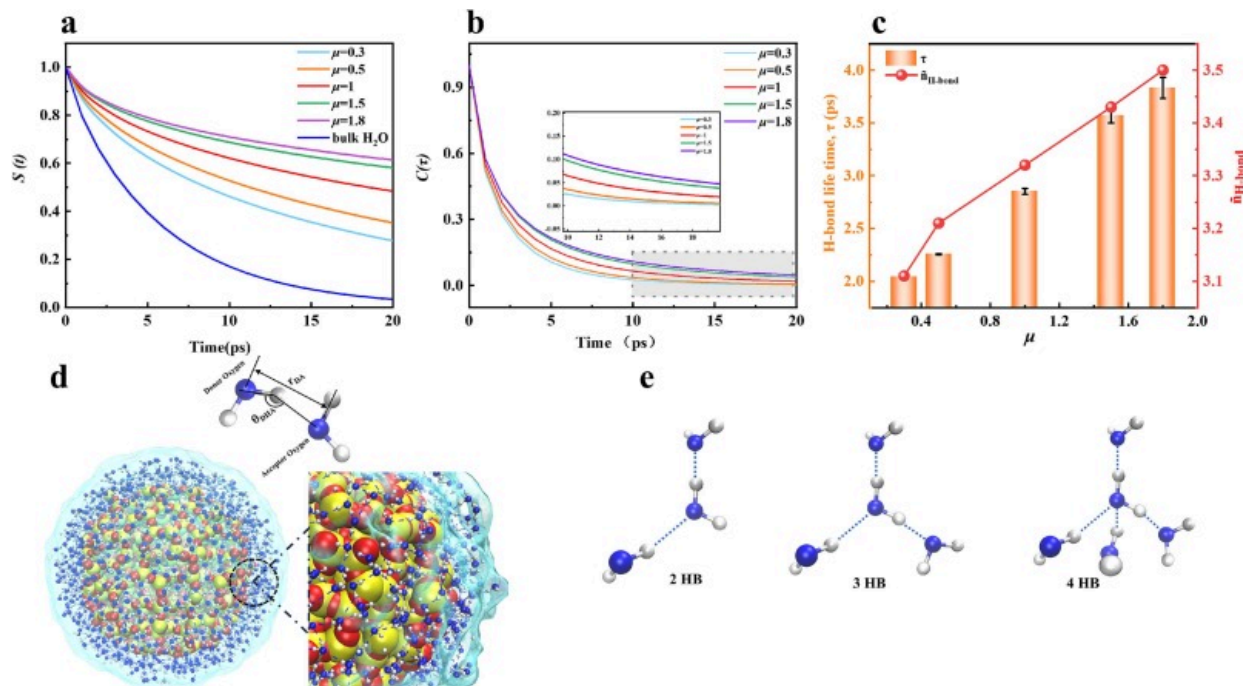
In this study, H-bonds are determined by geometric criteria, including the distance between hydrogen and acceptor atoms (H...A) and the donor-H-acceptor (D-H...A) plane angle. If (H...A) < 3.5 Å and (D-H...A) > 150°, it is considered that the H₂O molecule has formed an H-bonds. [70,71] The schematic diagram of the calculation model is shown in Fig. 8(d). The time autocorrelation function of the presence of the H-bond characterizes

the dynamic behavior of H-bond, with the calculation formula expressed as Eq. 9. [72]

$$C\tau = \frac{\sum h_{ij}t_0 h_{ij}t_0 + t}{\sum h_{ij}t_0^2} \quad (9)$$

where h_{ij} is utilized to indicate whether a H-bond has been formed between the paired atoms ij , with $h_{ij}=1$ denoting the formation of a H-bond and $h_{ij}=0$ indicating the absence of such a bond. [73] The appropriate value for t is determined through testing various t values and is based on the typical lifetime of water molecules under normal conditions. The calculation formula for the lifetime of H-bonds, denoted as τ , is provided in Eq. 10. [60]

$$\tau_x = \int_0^\infty C_\tau t dt \quad (10)$$



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Fig. 8. The interface of a-SiO₂ nanoparticles under different wettability (a) the survival probability of water molecules and bulk water (b) temporal autocorrelation function; (c) the average number of H-bonds and H-bonds lifetime formed by water molecules; (d) H-bonds network diagram at the interface of a-SiO₂ nanoparticles; (e) H-bonds structure formed by water molecules; (e) schematic diagram of H-bonds mode of water molecules [34].

Prior to delving into the kinetic characteristics of water molecules within the “solid-like liquid layer” at the nanoparticles interface, it is imperative to ascertain the temporal scale of water molecule residence at the particle interface, as the local translational and rotational mobility of water molecules is highly correlated with the duration they may remain within a given region. [60] Consequently, the lifespan of water molecules within the adsorbed layer of a-SiO₂ nanoparticles is determined by calculating the survival probability of water molecules. The computational domain for survival probability aligns with that for the RDF, with the results depicted in Fig. 8(b). The function $S(t)$ represents $(S(t))$ whether water molecules within a specified region at time $t=0$ retain their position in the region at a later time $t>0$. The decay rate of $S(t)$ is contingent upon the size of the region and the mobility of the water molecules within it; given the consistent computational domain, the magnitude of $S(t)$ is solely a function of the water molecule's mobility. Fig. 8(b) illustrates that the $S(t)$ of water molecules within the “solid-like liquid layer” is notably sensitive to changes in wettability. For weak wettability a-SiO₂ nanoparticles, $S(t)$ decreases by over 50%, while under conditions of strong wettability, the decline in $S(t)$ is less pronounced. Conversely, in the bulk water region, the $S(t)$ of water molecules converges rapidly in a very short span, indicative of the swift migration of water molecules. This difference is largely due to the fact that water molecules are not fixed in the same area, but move constantly. The increase of wettability weakens the mobility of water molecules at the interface of nanoparticles. The residence time of water molecules at the nanoparticles interface is much longer than that in the bulk water area, indicating that the water molecules at the nanoparticles interface are frozen and the

possibility of diffusion is further reduced. However, in general, the water molecules near the nanoparticles maintain a small mobility within 20ps.

The H-bonds between water molecules are dynamic and constantly recombine and break, possibly breaking at a moment's notice and recombining with other water molecules at a later moment. These dynamic changes in H-bonds lead to fluctuations in the total system's dipole moment. Therefore, according to the theory of fluctuation dissipation, the speed of the dynamic change of H-bonds will directly affect the stability of the H-bonds thermal network structure. The average lifetime of the H-bonds serves as a metric for the rate of these dynamic alterations, as determined by the time-correlation function($C(t)$), as illustrated in Fig. 8(b). The $C(t)$ of different wettability decays exponentially, but the decay rate is different. As the wettability increases, the decay rate of $C(t)$ gradually slows down. The H-bonds lifetime calculated from Eq.11 is shown in Fig. 8(c). The calculated value of the H-bonds lifetime for free water is $\tau = 1.61 \pm 0.02$, which is basically consistent with the 1.66 reported in the literature. [74] Overall, the H-bonds lifetime within the “solid-like liquid layer” is longer than that in bulk water, indicating a distinct molecular structure. The H-bonds lifetime increases with the increase of wettability, indicating that wettability has a significant effect on the stability of the H-bonds network of the “solid-like liquid layer”. The longer lifetime of H-bonds indicates that after they break, water molecules can promptly form new H-bonds with other water molecules. As more water molecules become linked by H-bonds and gather at the nanoparticles interface, the layered structure of water at the particle interface is further reinforced, resulting in a more stable and dense “solid-like liquid layer”. This is highly consistent with the distribution analysis of water density. Areas with higher water density facilitate the formation of more H-bonds, thereby contributing to structural stability and promoting the formation of a network for heat conduction through H-bonds, consequently enhancing the likelihood of heat transfer. These explanations align with the conclusions of Milischuk et al. [60] regarding the diffusion behavior of water molecules in silica nanopores. The prevalence of the H-bonds, particularly at the particle interface due to varying wettability, retards the exchange between surface water and bulk water,

increasing the residence time of surface water molecules. Consequently, the binding within the interface water molecules is stronger than that in bulk water, reducing their mobility and increasing H-bonds lifetimes.

Fig. 8(c) illustrates the average number of H-bonds that can be formed by water molecules within the “solid-like liquid layer”, with each H-bonds configuration corresponding to a unique pattern. It is observed that, under normal wettability, the average number of H-bonds per water molecule is approximately 3.18, increasing to 4 under strong wettability. According to relevant research, the maximum number of H-bonds that each water molecule can form is four, that is, a water molecule can form H-bonds with 4 surrounding water molecules. This tetrahedral H-bonds water structure is a common structural form of solid ice, and 2-hydrogen-bonded water is the most common form at room temperature and pressure. [32,34] The various H-bonds configurations are depicted in Fig. 8(d). It is worth noting that H-bonds is a weak interaction. The H-bonds can make water molecules in a more stable state. In liquid water, water molecules are connected to a dynamic H-bonds network. Due to the vibration of hydrogen atoms, the linear structure of H-bonds is destroyed. Therefore, an increase in wettability promotes the formation of a greater number of average H-bonds, inducing a structural transition akin to that of solid ice, as previously analyzed by the RDF. A higher average number of H-bonds translates to stronger intermolecular bonds, extending the H-bonds lifetime and reinforcing the layered structure of the water molecules at the particle interface. This stability enhances the formation of a hydrogen bond-mediated thermal conduction network, as evidenced in Fig. 8(c).

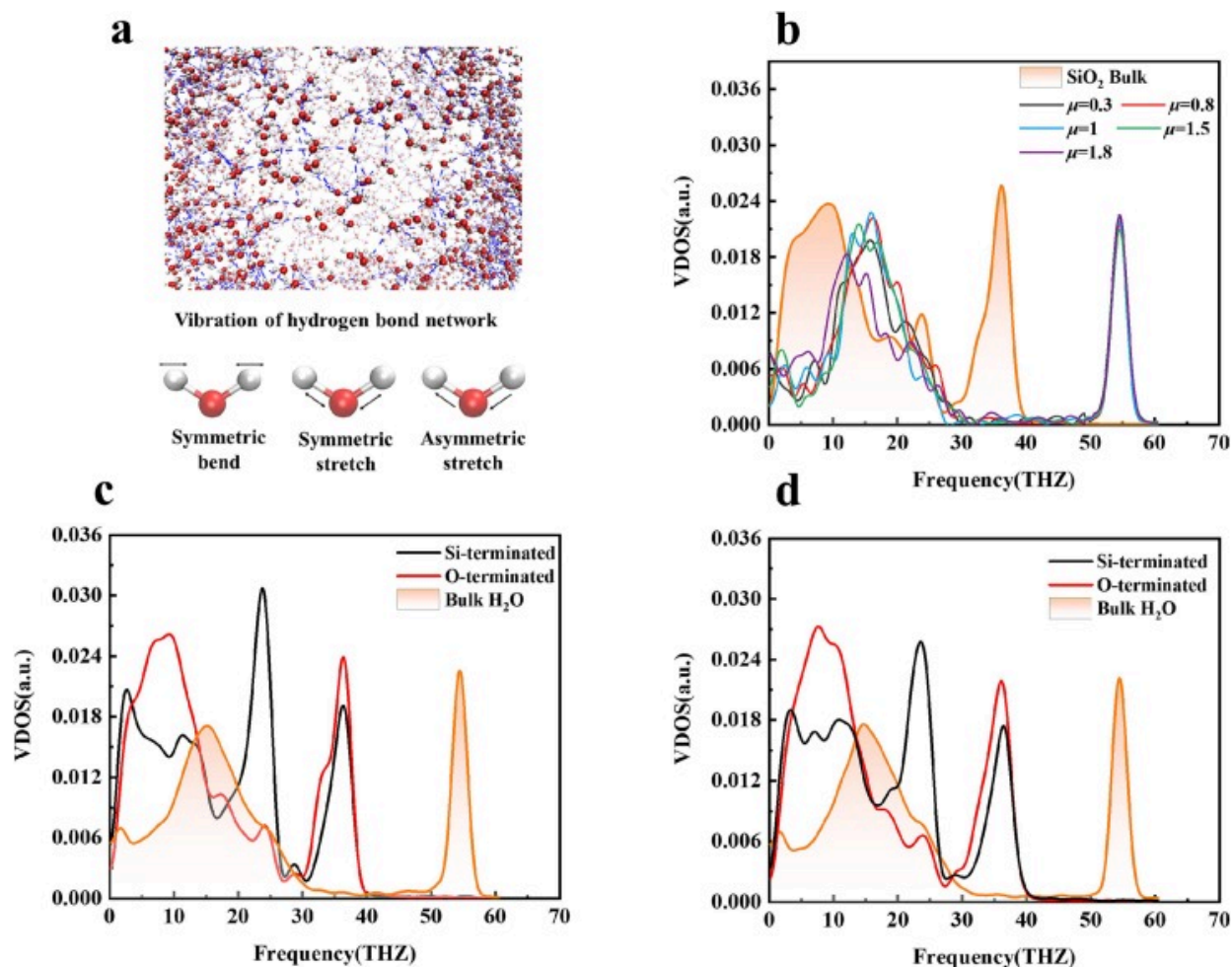
3.4. VDOS of water molecules at solid-liquid interface

The VDOS calculated from the velocity autocorrelation gives the distribution of the simple harmonic vibration modes in the system. Prior analyses have revealed that at the α -SiO₂ nanoparticles interface, water molecules organize into a layered structure resembling a solid, with their arrangement being orderly due to H-bonds interactions, placing this structure in an intermediate phase between the solid and liquid (ice and water) states.

[75] Within this configuration, the mobility of water molecules is akin to vibrational behavior, with energy transfer occurring through a mechanism akin to phonon transport. The VDOS offers a window into the structural state of the system and the scope and distribution of vibrational energy carriers. The distribution of the VDOS at the interface can be analyzed to better understand the microscopic mechanism of interfacial thermal transport. The computational region for VDOS at the interface is maintained consistent with that previously used for the RDF calculations.

The VDOS spectrum of a typical water molecule is characterized by three distinct absorption peaks, as illustrated in Fig. 9(a). The first peak, observed in the range of 5–30 THz, is associated with the vibrational modes of the hydrogen-bond network. The second peak, located between 50–60 THz, corresponds to the bending vibrations of internal bond angles within water molecules. The third peak, positioned at 90–105 THz, represents the stretching vibrations of intramolecular bonds. Recent investigations have demonstrated that different vibrational modes in the VDOS spectrum of water exhibit distinct influences on thermal effects, with the low-frequency region being identified as the predominant frequency range for thermal transport. [76] This observation has been consistently corroborated in our previous studies. [77] The VDOS spectra of water molecules and particles within the “solid-like liquid layer” at the silica interface under different wettability conditions are presented in Fig. 9(b). Analysis of these spectra reveals that the second peak in the high-frequency region remains relatively unaffected by changes in wettability, suggesting that the bending vibrations of water molecules are minimally influenced by surface wettability. However, significant frequency shifts are observed in the low-frequency region, primarily attributed to modifications in the H-bonds network vibrations. Specifically, as wettability increases, a pronounced frequency shift in the VDOS of water molecules within the “solid-like liquid layer” is observed in the low-frequency direction, particularly within the 5–15 THz range, which demonstrates substantial spectral overlap with the vibrational heat transfer spectrum of silica. This enhanced wettability has been found to significantly strengthen the vibrational coupling between water molecules and silica within the “solid-like liquid layer”. The intensified

vibrational coupling provides additional pathways for thermal transport, thereby amplifying vibrational coupling thermal transport within the “solid-like liquid layer” and promoting interfacial thermal transport, ultimately resulting in increased interfacial thermal conductivity. Previous structural analyses of water molecules and H-bond dynamics have revealed that enhanced wettability leads to increased adsorption of water molecules on the silica surface, prolonging their residence time and inducing structural transformations. [25,50] These transformations are associated with extended H-bonds lifetimes and reinforced intermolecular structures, establishing vibrational heat transfer as the dominant mechanism within the interfacial layer. This vibrational behavior has been shown to be intrinsically linked to the reorganization of the H-bonds network.



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Fig. 9. (a) Vibration mode of terahertz absorption spectrum of water [78] (b) the VDOS of water molecules in the "solid liquid layer" at the interface of a-SiO₂ nanoparticles under different wettability (c) the VDOS of atoms and bulk water at the interface of a-SiO₂ nanoparticles under weak wettability ($\mu=0.3$); (d) the VDOS of atoms at the interface of a-SiO₂ nanoparticles and bulk water under strong wettability ($\mu=1.8$).

Additionally, we analyzed the VDOS overlap between atoms at the a-SiO₂ nanoparticles interface and water molecules under strong and weak wettability, as shown in Fig. 9(c-d). It is observed that as wettability increases, there is no significant shift in the VDOS frequencies of different interface atoms, suggesting that the influence of surface atoms is not significant. This is consistent with the research on the ordering of water at the silica-water interface. [79] Therefore, the primary difference in the VDOS of water molecules that remain on the particle surface for an extended period is observed in the low-frequency range due to the stretching vibrations of H-bonds.

4. Conclusion

This study utilized molecular dynamics simulations to investigate the interfacial thermal transport mechanism between silica nanoparticles and water under varying wettability, with a particular focus on the formation of a “solid-like liquid layer” at the interface and its impact on interfacial thermal transport.

- (1) As wettability increases, the temperature jump at the silica nanoparticle-water interface gradually diminishes, and the interfacial thermal conductance significantly rises. The formation of a “solid-like liquid layer” at the particle interface is the primary cause. Within this layer, water molecules are restricted due to H-bonds interactions, leading to a decrease in their mobility and an extension of their residence time, thereby enhancing the interfacial thermal transport capacity.
- (2) With the increase in wettability, the number of H-bonds between water molecules in the “solid-like liquid layer” increases, and the lifetime of these H-bonds is prolonged. This promotes the adsorption of more water molecules onto the silica interface, forming an enhanced H-bonds network that provides additional pathways for heat transfer, further augmenting the interfacial thermal transport.

- (3) Analysis through the VDOS reveals that the low-frequency H-bonds stretching vibration modes of water molecules in the “solid-like liquid layer” closely coincide with the vibrational spectrum of silica, particularly in the low-frequency range of 5–15 THz.

CRediT authorship contribution statement

Ming Ma: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Xiaohui Zhang:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Can Xiong:** Writing – original draft. **Xiaoyan Huang:** Formal analysis. **Luyang Chen:** Formal analysis. **Shan Qing:** Software, Funding acquisition. **Hua Wang:** Supervision.

Funding sources

This work was supported by Yunnan Fundamental Research Projects (NO. [202301AT070469](#)), Yunnan Major Scientific and Technological Projects (NO. [202202AG050002](#)).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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